

Request for criticism:
have we established that trademark filings measure
anything — innovation, business formation, or otherwise?

Eric Silver

May 2026

What I am asking you to do. Attack the construct validity of this body of work. I am not asking whether the regressions are coded correctly (assume they are; the code is reproducible) or whether the prose is polished. I am asking the one question that, if answered badly, makes everything downstream worthless: *does a trademark filing, and the ΔKL statistic I compute on its goods/services text, measure anything real — innovation, business formation, or even meaningful commercial activity — or am I measuring the drafting habits of trademark attorneys and calling it economics?*

I have written this to make your job easy. Below I state the claims, then I steelman the case that the whole edifice is measuring noise, then I lay out the thin evidence I have for validity, then I offer four reframings from strong to weak and ask you to tell me which (if any) survives. Be as brutal as the strongest version of the objection warrants. I would rather kill the “innovation” framing now than have a referee or a serious reader kill it later.

1 The claims, briefly

There are two bodies of work sharing one measurement apparatus.

(A) The ΔKL papers (main.pdf, short.pdf). For each of 10,060,073 “clean” US trademark filings (1990–2021 cohorts), I compute a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ on the goods/services description text, against per-day exponentially decay-weighted reference distributions of the same NICE class (half-life $H = 2y$). $\Delta KL > 0$ when a filing’s vocabulary is unusual against the recent past and close to the near future — the directional resonance signal of Murdock, Allen, & DeDeo (2017), applied for the first time to commercially-incentivised text. The papers claim: ΔKL is an innovation measure; it predicts post-registration survival (U -shaped) and financial outcomes; it is statistically independent of patent counts ($r = +0.01$ on 174,569 firm-years) and of comparable correlation magnitude; the innovation tail aligns with the empirical content of optimal distinctiveness and the harvest tail is an unexplained surprise.

(B) The dynamism / ethnic-cluster work (dynamism.pdf, ethnic_clusters_note.pdf). Treating a firm’s first-ever filing (“debut”) as a proxy for business formation, I show trademark debuts cover 8–18% of Census High-Propensity Business Applications; I impute applicant race/ethnicity from surnames; I document ethnic-cluster industries, a Black trademark-rise / patent-fall divergence, and the collapse of an apparent “patent guild” signal once foreign inventors are stripped.

Everything in (B) inherits the validity of the measurement in (A), plus its own surname-imputation problems. So the load-bearing question is shared.

2 Steelman: the case that this measures nothing real

Here is the argument I most want you to either complete or refute. I find it uncomfortably strong. [moderate] that at least the first three links are serious and not fully answered by the current drafts.

1. **A trademark is a brand, not an invention, and the goods/services text is a legal instrument, not a description of novelty.** The g/s description exists to define the *scope of protection*, drafted by an attorney (or selected from the USPTO’s pre-approved *ID Manual*) to be as broad as the examiner will allow. Two firms selling physically identical products can write wildly different descriptions; a firm with a genuinely novel product can file boilerplate. So ΔKL measures the novelty of the *prose*, and the prose is several steps removed from the product. Why should I believe the prose tracks the innovation?
2. **Vocabulary novelty has at least four generators, and I have separated none of them.** A high- ΔKL filing uses words that are unusual-then-common. That could be (a) a genuine new product category (crypto, vape, NFT); (b) legal-drafting fashion; (c) attorneys copying each other’s successful protective language; (d) the USPTO updating its ID Manual, which mechanically shifts the reference vocabulary for everyone. The cannabis/crypto examples I parade are consistent with (a), but they are also exactly the cases where new ID-Manual entries appeared. I have not shown the signal is (a) rather than (d).
3. **The resonance interpretation was validated on a different species of text.** Murdock, Allen, & DeDeo (2017, *Cognition*) developed the framework on Darwin’s private reading notebooks; Barron, Huang, Spang, & DeDeo (2018, *PNAS*) on French Revolution parliamentary debates — in both, the text is a window into the author’s cognitive or rhetorical state, authored by the person whose novelty is being measured. A trademark g/s description is drafted for protection scope by a third-party agent. Importing the “resonance = the field moved toward this filing” interpretation assumes the signal means the same thing in a corpus where the author is a lawyer optimising a legal outcome. I assert the analogy; I do not re-validate it.
4. **Selection: I only ever observe the $\sim 8\text{--}18\%$ of business starts that trademark.** Coverage against Census High-Propensity Business Applications is 8% (2004–08) rising to 18% (2019–24). The trademark-filing population is selected on branding intensity, consumer-facing orientation, and capitalisation. Every sentence I write “about firms” is really about the trademark-filing subset, and that subset is exactly the set of firms for which branding-vocabulary is salient — which is precisely the variable ΔKL measures. The selection is on the regressor.
5. **“Survival” is trademark maintenance, not firm survival.** My headline outcome — 5-year survival — is the owner paying the § 8 declaration-of-use fee. A live firm can abandon a mark; a dead firm’s mark can sit registered until renewal lapses. The outcome is a portfolio-management decision correlated with, but not identical to, firm viability. The *U*-shape on ΔKL could be a *U*-shape in *which marks firms bother to maintain*, not in which firms survive.
6. **The financial validation is small and one of the numbers is implausible.** The patent R^2 decomposition is on 8,118 firm-years; the gross-margin effect is +3.2pp per σ (SE 0.6); these

are real but modest. The claimed +28 percentage-point 4-year excess return per σ of debut ΔKL is, on its face, [low] confidence and probably too large to believe — a σ move in a text statistic should not move 4-year returns by 28pp; I suspect a specification or sample-composition artefact and want it stress-tested.

7. **Independence from patents cuts both ways.** I sell $r = +0.01$ with patent counts as evidence that ΔKL is “a distinct innovation channel.” The skeptic’s reading is the opposite: ΔKL is uncorrelated with the one measure we *know* tracks invention because ΔKL is not measuring invention at all — it is measuring marketing/branding activity, which is orthogonal to R&D by construction. Independence is necessary for “distinct channel” but equally consistent with “not an innovation measure.”
8. **The ethnic / dynamism layer multiplies the problem.** Surname imputation is probabilistic, restricted to the $\sim 32\%$ of filers parsing as individuals, static to the 2010 Census, and weakest exactly for Black surnames (Smith/Johnson/Williams load on both White and Black). On top of that it inherits the whole proxy-validity question: if trademark debut is not business formation, the coverage ratios and ethnic-composition trends describe “who files trademarks,” not “who starts businesses.”

3 What I actually have for validity (honest accounting)

I am not claiming the steelman wins. Here is the evidence on the other side, with its limits stated.

- **Patent independence + comparable magnitude.** $r = +0.01$ with patent counts (174,569 firm-years); incremental 4-year-return R^2 of +0.0046 for ΔKL vs +0.0024 for $\log(1 + \text{patents})$, roughly additive. *Limit:* small R^2 absolute; comparable-to-patents is a low bar if patents themselves explain little return variance in this panel.
- **Survival is U-shaped under multiple operationalisations and in 32/44 NICE classes.** *Limit:* all of them are maintenance-based; the robustness is across definitions of maintenance, not across independent firm-outcome data.
- **Cluster-viability mediation absorbs 84% of the quadratic.** This is the one result that looks mechanism-like rather than mechanical. *Limit:* the mediator is itself built from trademark data (cosine-neighbour survival), so it does not break out of the corpus.
- **Coverage brackets Dinlersoz et al. (2018).** Our 8–18% spans their restricted-LBD 10–15%. *Limit:* agreement on the *coverage fraction* is not validation that the covered filings *mean* anything; it just says we count debut filers the way Census counts them.
- **Face validity.** Canonical winners skew positive ΔKL ; flux-neutral tokens (organisational, conformity) die. *Limit:* selected ex-post; not a test.

My own read [moderate]: I can defend “ ΔKL is a nontrivial, patent-orthogonal signal that predicts trademark-level outcomes and correlates weakly with firm financials.” I cannot currently defend “ ΔKL is an innovation measure” against a determined skeptic, because every outcome I validate against is either inside the trademark corpus or a small noisy financial panel.

4 Four reframings — which survives?

I want you to tell me which of these I can defend, and whether the strong claim is salvageable with additional work or should be abandoned.

(I) Strong — “ ΔKL is an innovation measure.” The current framing. Requires that vocabulary novelty in g/s prose tracks product/market innovation. My confidence this survives review as stated: [low].

(II) Medium — “ ΔKL is a leading indicator of commercial category emergence.” Claims the corpus-level signal (which *vocabularies* are forming and being adopted) is real and useful, while staying agnostic about whether any individual filer is “innovative.” The cannabis/crypto/NFT waves are the evidence; the firm-level survival/returns become secondary. Confidence: [moderate].

(III) Weak/descriptive — “ ΔKL measures the linguistic position of a filing relative to its NICE class.” Purely descriptive. Every downstream correlation (survival, returns, ethnic clusters) becomes an empirical regularity to be explained, not a validation of an innovation construct. This is almost certainly defensible [high] but gives up the headline.

(IV) Activity proxy — “a trademark debut is formal commercial activity with branding intent.” For the dynamism work specifically: not innovation, not business formation *per se*, but a measurable, name-disclosed, near-real-time signal of branding-intent commercial activity, explicitly the 8–18% branding-intensive slice of formation. Confidence: [moderate].

A plausible restructure: lead with (III) as the measurement contribution, establish (II) at the corpus level with the category-emergence evidence, and present the firm-level innovation/financial results as *suggestive regularities under (III)* rather than as proof of (I). The dynamism work then stands on (IV) and does not need (I) at all. Tell me if this is cowardice or rigor.

5 Specific questions

1. Is link 1 (prose $\neq$ product) fatal, or is there a defensible argument that attorney-drafted scope language still carries firm-level signal because the firm chooses what to claim?
2. What is the cheapest decisive test that would separate “genuine category emergence” from “ID-Manual / drafting-fashion drift”? (My best idea: hold out the ID-Manual update history and show ΔKL predicts adoption *before* the manual codifies the term. I have not done this.)
3. Is there any external, non-trademark, non-financial outcome I could validate survival against — business registration death, UCC-lien termination, Form-D withdrawal, domain expiry — that would break the maintenance-vs-survival circularity?
4. Does the $+28pp/\sigma$ return number kill my credibility on sight? Should I drop it pending re-estimation?
5. For the ethnic work: is surname imputation usable at all given the Black-surname cross-loading, or should I restrict to the Hispanic and API analyses where the Census classifier is discriminating and drop Black claims to a flagged-caveat appendix?
6. Is reframing (III)+(II)+(IV) the right retreat, or is it giving up a defensible (I) too cheaply?

6 Artifacts and reproduction

All on branch `business-dynamism` of the project repo. `main.pdf` (long ΔKL paper, 223pp), `short.pdf` (7pp), `dynamism.pdf` (13pp), `ethnic_clusters_note.pdf` (21pp). Pipelines: `Makefile` and `Makefile.dynamism`. Every figure regenerates from `scripts/` against the per-NICE-class parquets; the surname work uses the 2010 Census surname file and a full re-parse of the USPTO XML for applicant country (`owner_address.parquet`, 13,993,768 records).

References

- Akcigit, U., & Ates, S. T. (2021). Ten facts on declining business dynamism and lessons from endogenous growth theory. *American Economic Journal: Macroeconomics*, 13(1), 257–298.
- Barron, A. T. J., Huang, J., Spang, R. L., & DeDeo, S. (2018). Individuals, institutions, and innovation in the debates of the French Revolution. *PNAS*, 115(18), 4607–4612.
- Dinlersoz, E., Goldschlag, N., Myers, A., & Zolas, N. (2018). An anatomy of US firms seeking trademark registration. Center for Economic Studies WP 18–46, US Census Bureau.
- Decker, R. A., Haltiwanger, J., Jarmin, R. S., & Miranda, J. (2014). The role of entrepreneurship in US job creation and economic dynamism. *Journal of Economic Perspectives*, 28(3), 3–24.
- Murdock, J., Allen, C., & DeDeo, S. (2017). Exploration and exploitation of Victorian science in Darwin’s reading notebooks. *Cognition*, 159, 117–126.
- Word, D. L., Coleman, C. D., Nunziata, R., & Kominski, R. (2008). Demographic aspects of surnames from Census 2000. US Census Bureau working paper.